

Exam Aprendizaje Automático y Minería de Datos

Bachelor's Degree in Video Game Development, January 2026

General instructions

The exam consists of two parts, a theoretical part worth 2 points and a practical part worth 8 points.

The theoretical part must be submitted on campus with the student's name without spaces and in PDF format (you can use Word to export to PDF File>Export>Create PDF), Word or TXT format.

The practical part must be submitted in a Jupiter Notebook inside a zip file or, failing that, in a normal Python .py file.

The zip file must also include your library with the implementation of the generalised multilayer perceptron for multiple layers and neurons per layer. This library must be based on the code submitted in practicals 4 and 5, although it may have modifications.

If the code has nothing to do with that of practical 5, it will be considered cheating on the exam and you will fail the course.

The practical must be executable without the teacher's intervention; if it does not run, there will be a penalty in the grade.

For the practical and theoretical parts, you are allowed to use campus resources, notes and any support material you bring with you. You will not have access to the internet, only to the campus for the entire exam.

The use of ChatGPT or similar AI tools is not permitted, nor is communication between yourselves or with third parties. If detected, the practical will be considered copied and will receive a 0. Mobile phones are also not permitted during the exam.

Theoretical part (2 points)

Question 1 (0.25 points)

Given the following scenario: We have a model with a training accuracy of 90, a validation accuracy of 80, and a test accuracy (without intervention in training) of 70. Describe in detail what this data tells you about the model, the data, and its training.

Question 2 (0.25 points)

If we have a training accuracy of 90 and a validation accuracy of 70 in a model we have just trained, what actions would you take to improve the results? Explain your answer.

Question 3 (0.25 points)

If, during training, depending on the random state, the validation accuracy changes between values between 0.7 and 0.9, what could be happening and how would you fix it? What could be happening and how would you solve it?

Question 4 (0.25 points)

If we have these confusion matrices where the rows are the predicted values and the columns are the actual values of two models trained with the same dataset. What can we say about both models? In which scenarios would one model be preferable to the other? Explain your answer.

A	T	F
T	100	2
F	20	88

B	T	F
T	115	10
F	5	80

Question 5 (0.25 points)

We have trained a multilayer perceptron model with a dataset provided with 10 input parameters. After testing all possible hyperparameters of the model, we were unable to get the validation error to exceed the required baseline performance of 80%, obtaining a maximum of 77%, although the training error did exceed 80%, obtaining 87%. Given this situation, explain what measures you should take to obtain a model that could potentially exceed 80%.

Question 6 (0.25 points)

We have a set of images that we want to classify, and to do so we use a convolutional model composed of convolutional and fully connected layers. However, after several attempts to get it to learn correctly, our validation accuracy does not reach the required level, and we have detected that the training accuracy and the validation accuracy differ by 20% in favour of the training accuracy. What could we do to improve the result?

Question 7 (0.25 points)

What type of network would you use so that, based on a dataset of handwritten number images such as the one in practice 4 and the digit entered by the user via keyboard, the network is capable of generating an image of that number? Briefly explain how you would train it.

Question 8 (0.25 points)

We need to build a model that allows us to construct an agent capable of beating a rogue-like game in which the scenarios are procedurally generated. The agent must be able to beat the game, but it does not matter how long it takes to learn how to do so. What type of model would you use? Explain your answer.

Estimated time: 40 minutes, maximum time: 50 minutes.

Practical part (8 points)

Estimated time: 1 hour 50 minutes; maximum time: 2 hours 20 minutes.

We have the **Customer.csv** dataset attached to the exam instructions.

This dataset contains information about customer profiles:

Variable Definition

- ID: Unique ID
- Gender: Gender of the customer
- Ever_Married: Marital status of the customer
- Age: Age of the customer
- Graduated: Is the customer a graduate?
- Profession: Profession of the customer
- Work_Experience: Work Experience in years
- Spending_Score: Spending score of the customer
- Family_Size: Number of family members for the customer (including the customer)
- Var_1: Anonymised Category for the customer
- Segmentation: (target) Customer Segment of the customer

We want to build several models to study their behaviour in order to predict which segment a new customer belongs to.

Exercise 1 (1 point) Clean the dataset and perform the necessary transformations so that the data can be applied to any machine learning model. Justify the decisions you have made in a markdown cell. Apply some type of imputation to at least one attribute. Choose which one would make the most sense and justify it in markdown.

Exercise 2 (0.5 points): Graphically represent the data to show the distribution of the classes with a scatter plot where each class has a colour.

Exercise 3 (2 points): Use your Multilayer Perceptron from exercises 4 and 5 to build the prediction model. Calculate accuracy and the confusion matrix. Perform a partition with **random_state=13** and a test size of 20%. The minimum accuracy you must achieve is 50%.

The model can have any number of layers, but **there must be a test performed with at least three hidden layers**, even if it is not the final model you choose because it does not produce the best result. **Make it very clear which model you finally choose** if there are two versions.

Exercise 4 (1 point): Use SKLearn's MLPClassifier to train a model. This model must achieve at least 50% accuracy. The configuration may be different from that of the previous exercise.

Exercise 5 (1 point): Use the KNN algorithm to achieve 48% accuracy.

Exercise 5 (1 point): Use the RandomForestClassifier algorithm to achieve 53% accuracy.

Exercise 5 (0.5 points): Analyse the training and validation accuracy and explain the situation you detect based on your model's results and, if necessary, what measures you could take to improve the results.

Exercise 6 (1 point): Implement the option for hidden layers to use both relu and sigmoid as activation functions and the output layer to use both sigmoid and softmax. To do this, you must modify a copy of your multilayer perceptron called MLPRelu, in which the following fields must appear in the constructor:

```
"""
Constructor: Computes MLPRelu.
Args:
    inputLayer (int): size of input
    hiddenLayer (int): size of hidden layer.
    outputLayer (int): size of output layer
    seed (scalar): seed of the random numeric.
    epislom (scalar) : random initialization range. e.j: 1 = [-1..1], 2 = [-2,2]...
    function = "sigmoid" | "relu"
    out_function = "sigmoid" | "softmax"
"""
```

```
def __init__(self, inputLayer, hiddenLayer, outputLayer, seed=0, epislom = 0.12, function = "s
```

To prove that it works, train a model with Relu and softmax that gives a value similar to that obtained with sigmoidal.

The derivative of Relu is as follows:

$(a > 0) * 1$

When using Relu, cross entropy (cost) must be calculated with the xlogy function from scipy.special to prevent $\text{Log}(0)$ values from being generated, which produce NaN values.